A U-ADMISSIBLE BLOWUP MAKES THE STRICT TRANSFORM FLAT AND OF FINITE PRESENTATION, FOR AN OVER-S SCHEME FLAT OVER A QUASI-COMPACT OPEN U AND OF FINITE TYPE OVER S

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The engine of the proof is the flattening theorem of the same chapter, which is retained as a tagged navigation pointer rather than as a counted item; the reduction from the strict transform of the pushforward of the structure sheaf to the strict transform of X is written out in the counted span.

Editorial note. Canonical location of the counted proposition: Stacks tag 081R (sources/stacks-flat/pinned-original-source.tex line 9123).

Allowed standard prerequisites (editorial).

- The flattening theorem for a quasi-coherent module on a scheme of finite type over a quasi-compact quasi-separated base. Same chapter, Theorem labelled theorem-flatten-module (source line 9017), invoked at proof line 9158. Supplies the U-admissible blowup S' -> S making the strict transform of i_*O_X flat and of finite presentation; it is the engine of the proof, and the reduction that turns the conclusion about i_*O_X into the conclusion about X' is carried out in the counted span.
- Admissible blowups: dominating a finite family, and behaviour of strict transforms. Divisors, Lemma labelled divisors-lemma-dominate-admissible-blowups (Stacks tag 080N) with Remark labelled remark-successive-blowups (source line 8611), and Divisors, Lemma labelled divisors-lemma-strict-transform-affine (Stacks tag 080G), invoked at proof lines 9148, 9166. Supply the blowup dominating all the local solutions and the commutation of strict transform with pushforward along an affine morphism.
- An affine scheme of finite type over S embeds into an affine space over S. Morphisms, Lemma labelled morphisms-lemma-quasi-affine-finite-type-over-S (Stacks tag 04II), invoked at proof line 9154. Supplies the immersion j : X -> A^n_S and the quasi-compact open V in which j stays a closed immersion.

Source reference. Original lemma 080N: sources/stacks-chow/tags.txt, source lines 10188–10188. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9148 of the counted proof. It is

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Source reference. Original remark 080Y: sources/stacks-flat/pinned-original-source.tex, source lines 8611–8611. Same-chapter pointer into the author environment remark at source line 8611; invoked at proof line 9149 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 04II: sources/stacks-chow/tags.txt, source lines 5904–5904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9154 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 0815: sources/stacks-flat/pinned-original-source.tex, source lines 9017–9017. Same-chapter pointer into the author environment theorem at source line 9017; invoked at proof line 9158 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080G: sources/stacks-chow/tags.txt, source lines 10181–10181. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9166 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let X be a scheme over S. Let $U \subset S$ be a quasi-compact open. Assume*

(1) *$X \to S$ is of finite type and quasi-separated, and*
(2) *$X_U \to U$ is flat and locally of finite presentation.*

Then there exists a U-admissible blowup $S' \to S$ such that the strict transform of X is flat and of finite presentation over S'.

Proof. Since $X \to S$ is quasi-compact and quasi-separated by assumption, the strict transform of X with respect to a blowing up $S' \to S$ is also quasi-compact and quasi-separated. Hence to prove the lemma it suffices to find a U-admissible blowup such that the strict transform is flat and locally of finite presentation. Let $X = W_1 \cup \ldots \cup W_n$ be a finite affine open covering. If we can find a U-admissible blowup $S_i \to S$ such that the strict transform of W_i is flat and locally of finite presentation, then there exists a U-admissible blowing up $S' \to S$ dominating all $S_i \to S$ which does the job (see Divisors, Lemma 080N; see also Remark 080Y). Hence we may assume X is affine.

Assume X is affine. By Morphisms, Lemma 04II we can choose an immersion $j : X \to \mathbf{A}^n_S$ over S. Let $V \subset \mathbf{A}^n_S$ be a quasi-compact open subscheme such that j induces a closed immersion $i : X \to V$ over S. Apply Theorem 0815 to $V \to S$ and the quasi-coherent module $i_*\mathcal{O}_X$ to obtain a U-admissible blowup $S' \to S$ such that the strict transform of $i_*\mathcal{O}_X$ is flat over S' and of finite presentation over $\mathcal{O}_{V \times_S S'}$. Let X' be the strict transform of X with respect to $S' \to S$. Let $i' : X' \to V \times_S S'$ be the induced morphism. Since taking strict transform commutes with pushforward along affine morphisms (Divisors, Lemma 080G), we see that $i'_*\mathcal{O}_{X'}$ is flat over S and of finite presentation as a $\mathcal{O}_{V \times_S S'}$-module. This implies the lemma. □